Introduction to machine learning and ethics

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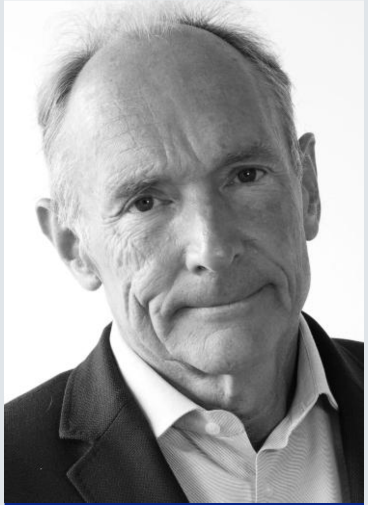
Founded in 2012, the Open Data Institute (ODI) is an international, independent and not-for-profit organisation based in London, UK.

Sir Nigel Shadbolt

Executive Chair and Co-founder of the ODI



Sir Tim Berners-Lee

President and Co-founder of the ODI

Our mission & Vision

We work with companies and governments to build an open, trustworthy data ecosystem.

We want people, organisations and communities to use data to make better decisions and be protected from any harmful impacts.

Enable you to apply machine learning and AI techniques to data and discover how ethical frameworks can help you avoid teaching your machines bad habits.

Introductions

Name?

What you do?

What brings you to the training today?

What did you major in (degree etc.)?

Schedule

09:30: Machine learning foundations: Exploratory data analysis

11:00: Break

11:15: Making inferences from data

12:00: The ML Classification Challenge

it is up to your group to decide when you take 45 minutes for lunch in this gap

13:45: The Results

14:15: Machine learning, ethics and fairness

15:00: Close

Machine learning is a subset of **artificial intelligence** in the field of **computer science** that often **uses statistical techniques** to give computers the ability to "learn" with data, without being explicitly programmed.

Wikipedia

Machine learning *allows computers to* "learn" with data, without being explicitly *told the answers*.

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Wikipedia

Machine learning isn’t magic.

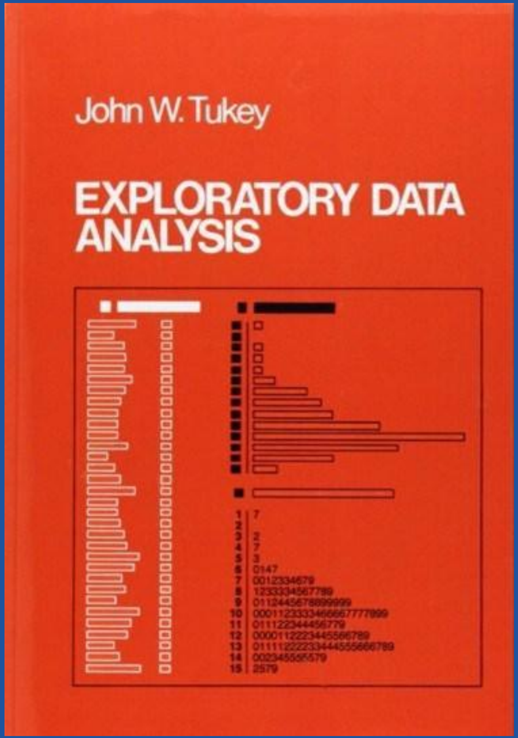
It’s basically statistics, logic and rules at scale.

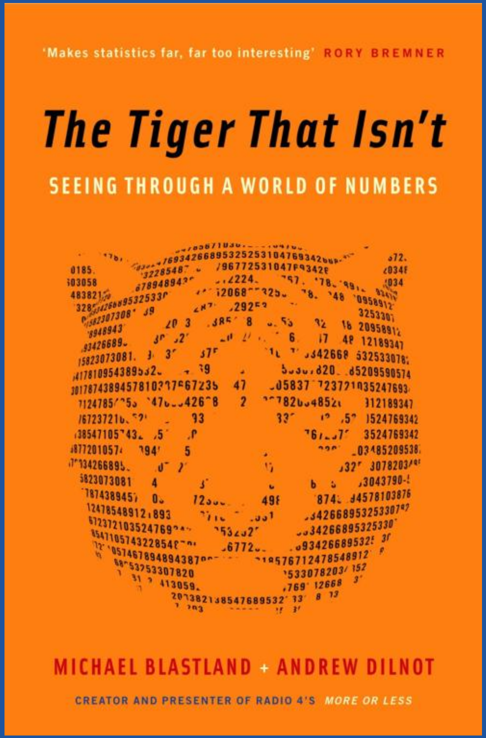
The machine can help us find patterns in data.

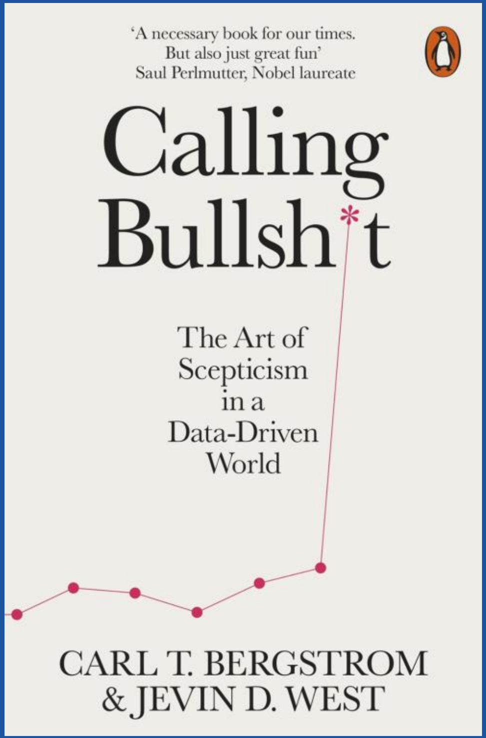
It can then use those new patterns it has learnt to predict or create new things.

All of it is made by humans, who are both amazingly intelligent and stupid all at the same time.

Further reading







Activity 1

Can you tell me what the following dataset represents
https://training.theodi.org/number-cards

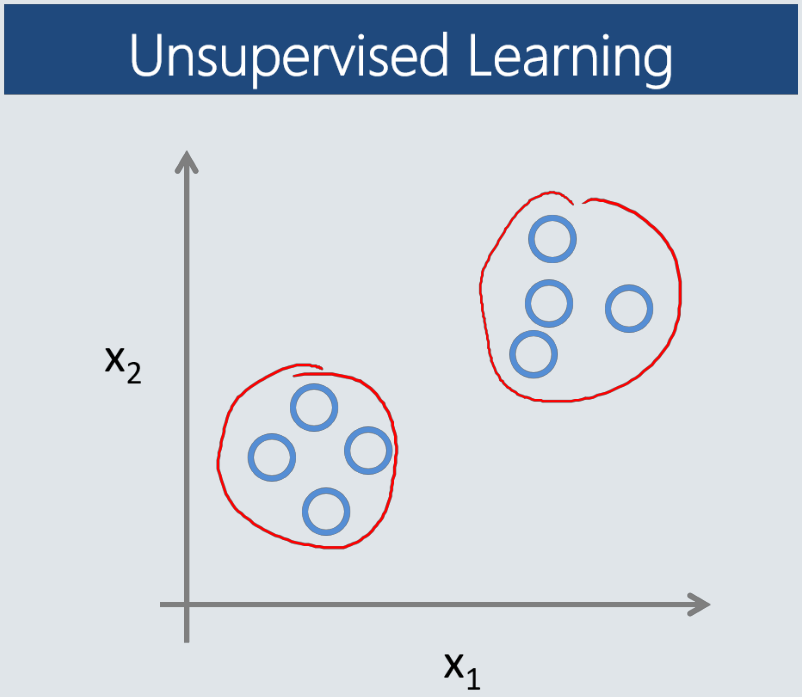
Clues
1. The dataset is complete
2. There is no rounding in the numbers
3. The numbers are discrete (none are additions of others)

Steps
1. What facts can you tell me about the data?
2. If there were 24,000 datapoints not 24, what summary statistics would you use to give you an indication of the shape and insight from the data?
3. Given (1) and (2), what does this data represent?

Unsupervised learning

It is used for clustering population in different groups, which is widely used for segmenting customers in different groups for specific intervention.

Examples of Unsupervised Learning:
Apriori algorithm, K-means.



Averages

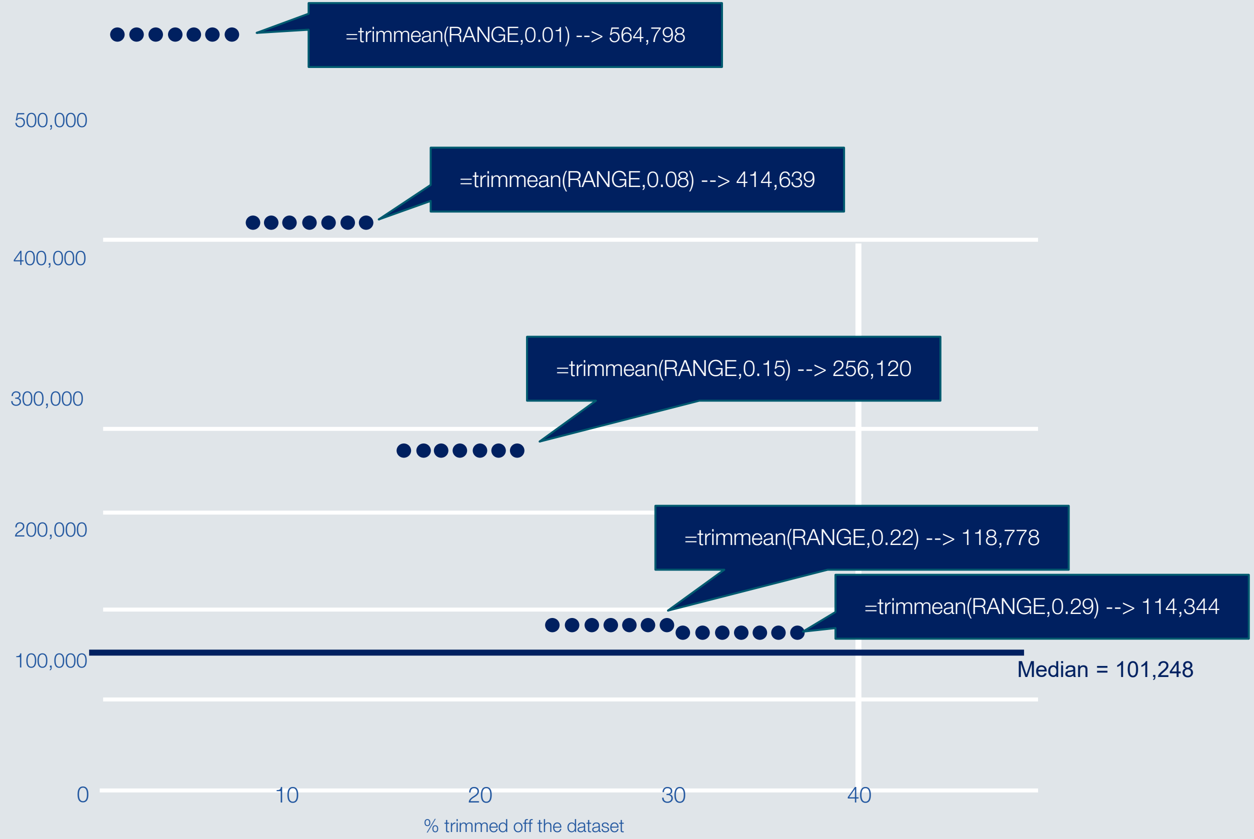
33750
33750
33750
33750
44000
44000 ← Mode (44,000)
44000
44000
44000
60000
65000
75000
80000
90000 ← Median (~101,248)
112496
120000
125000
145000
170000
174375
185000
190000
200000
275000
292499 ← Mean (564,798)
3500000
4600000
5000000

Outliers

[redacted]

44000
44000
44000
44000
44000
60000
65000
75000
80000
90000
112496
120000
125000
145000
170000
174375
185000
190000
200000
275000

[redacted]

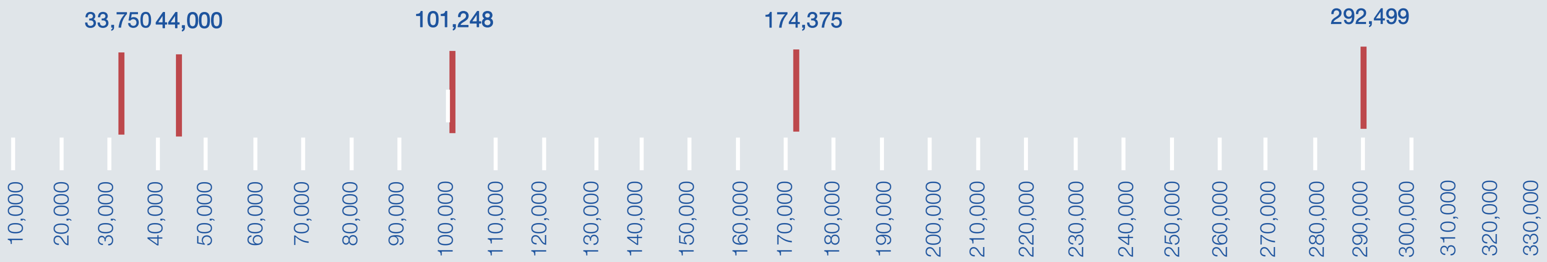


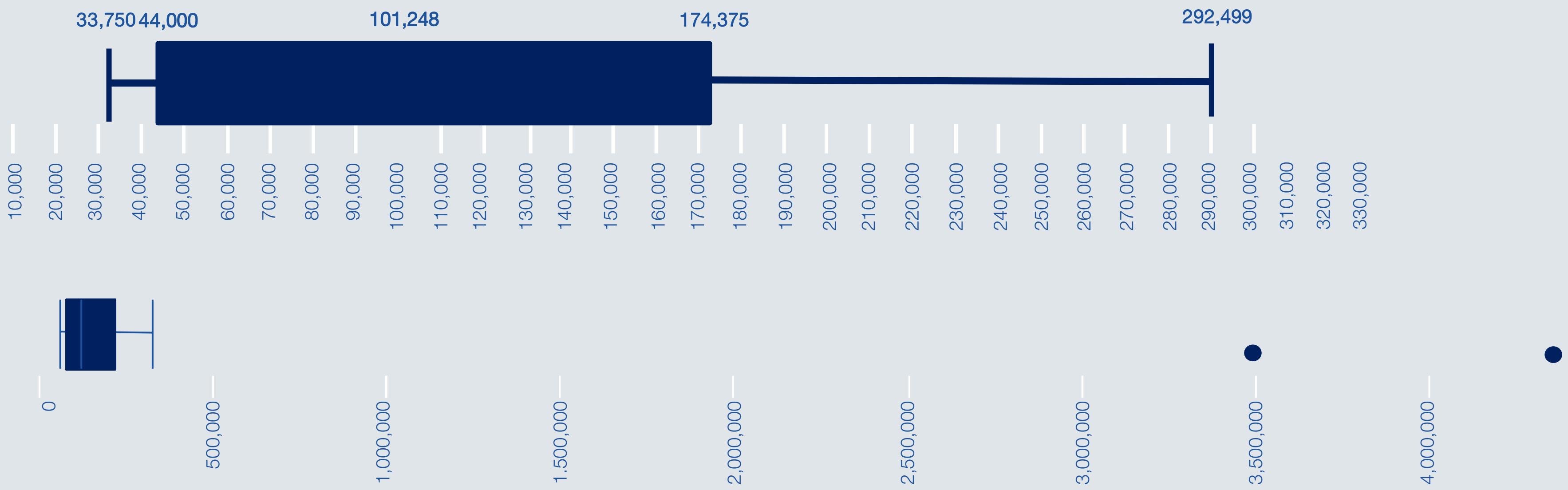
Formula to calculate a trimmed mean: =trimmean(RANGE,percent)

5-number summary: when mean or median isn't enough

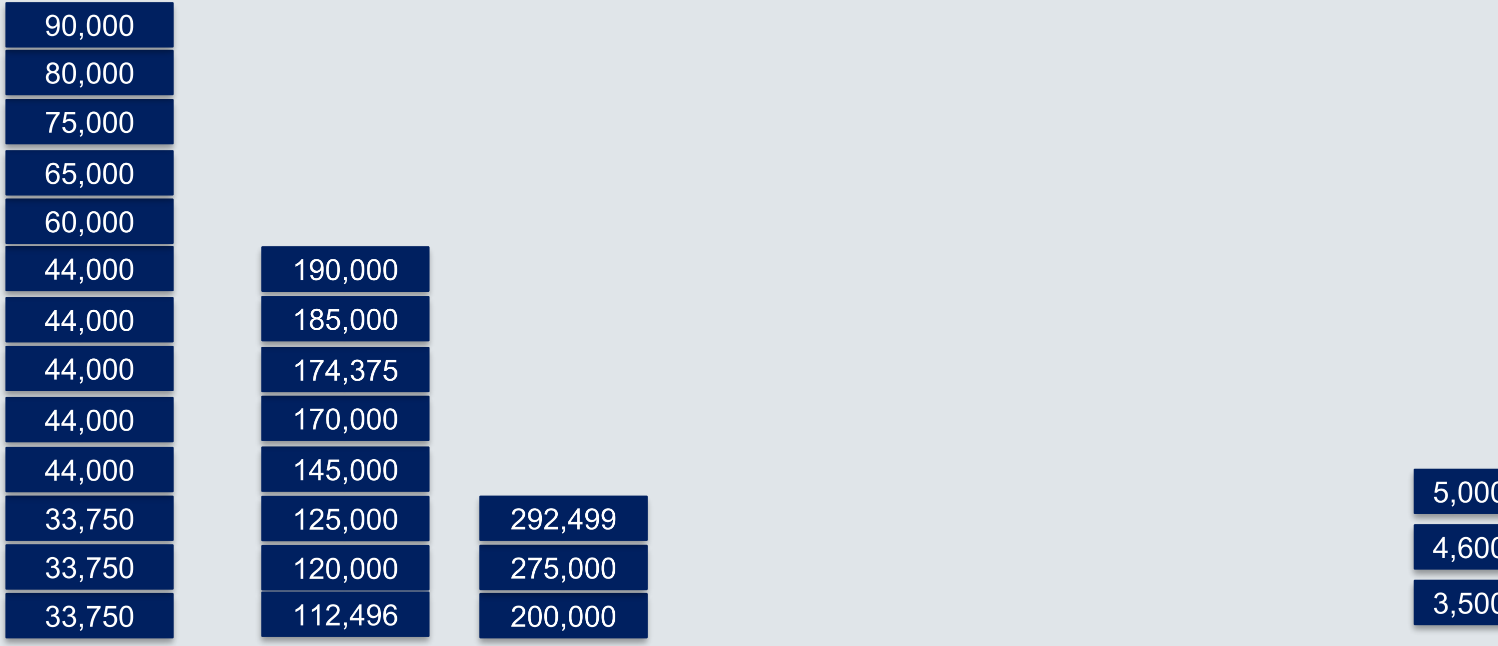
~~33,750~~	~~33,750~~	~~33,750~~	33,750	44,000	44,000
44,000	44,000	44,000	60,000	65,000	75,000
80,000	90,000	112,496	120,000	125,000	145,000
170,000	174,375	185,000	190,000	200,000	275,000
292,499	~~3,500,000~~	~~4,600,000~~	~~5,000,000~~		

- Minimum value
- Maximum value
- Median
- Lower quartile
- Upper quartile

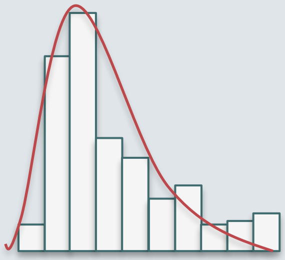



33,750
44,000
101,248
174,375
292,499
10,000
20,000
30,000
40,000
50,000
60,000
70,000
80,000
90,000
100,000
110,000
120,000
130,000
140,000
150,000
160,000
170,000
180,000
190,000
200,000
210,000
220,000
230,000
240,000
250,000
260,000
270,000
280,000
290,000
300,000
310,000
320,000
330,000
0
500,000
1,000,000
1,500,000
2,000,000
2,500,000
3,000,000
3,500,000
4,000,000

Shape of data

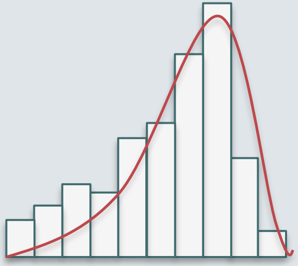


Distributions



Positively Skewed

(mean is *skewed* **higher** than median)

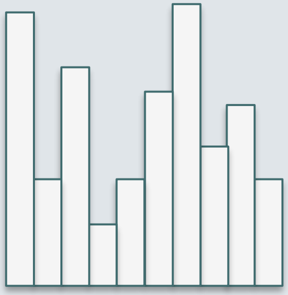


Negatively Skewed

(mean is *skewed* **lower** than median)

Normal distribution

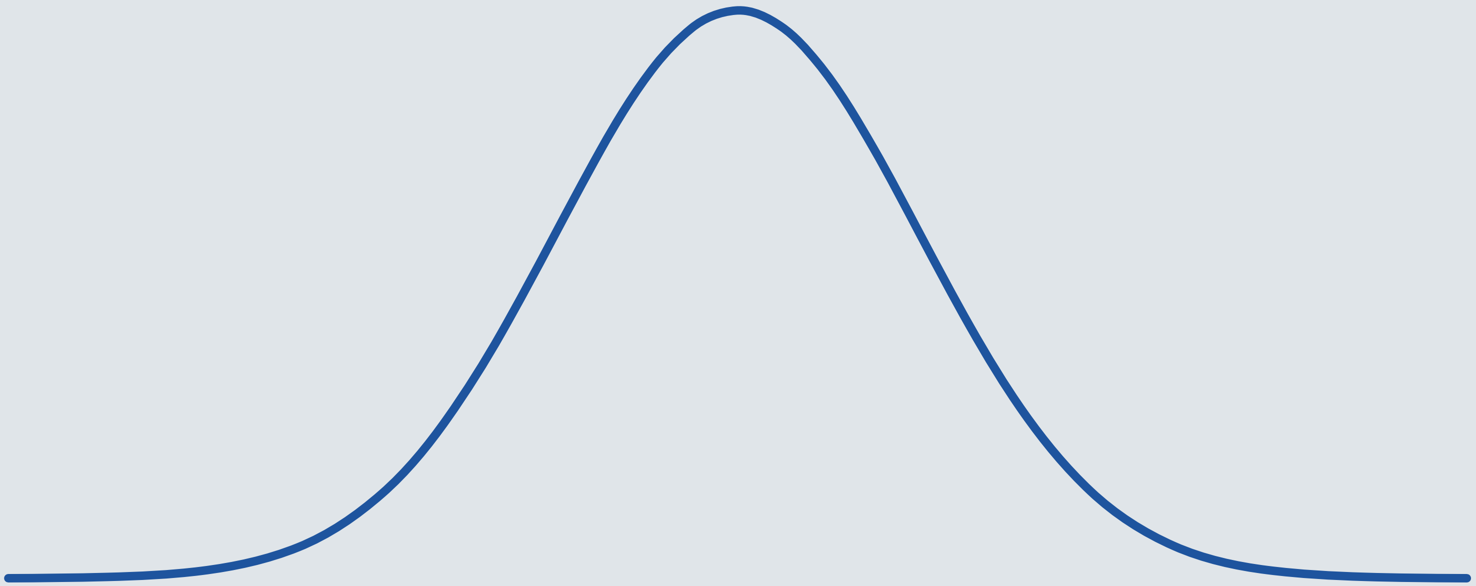
(aka. Gaussian Distribution, Bell Curve)



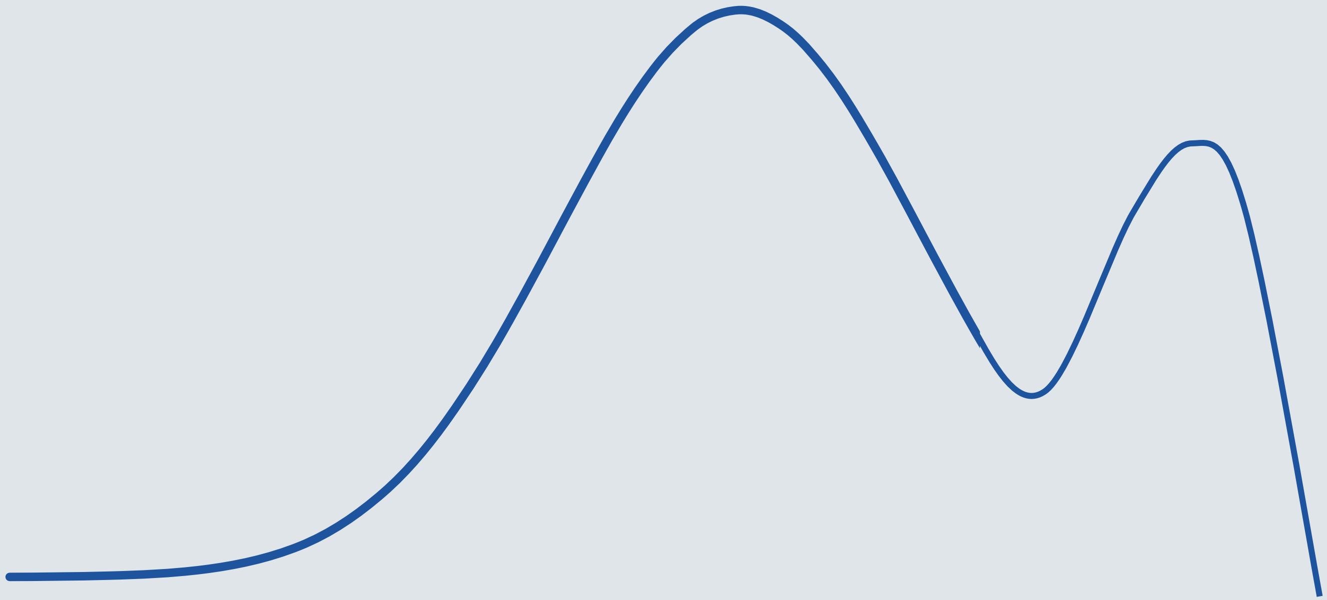
Multimodal distribution

(many modes)

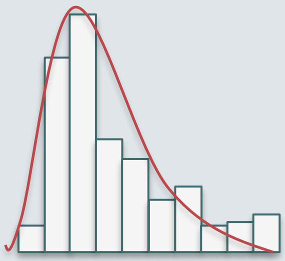
Histograms and targets



Histograms and targets



Distributions



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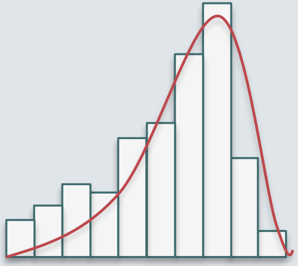

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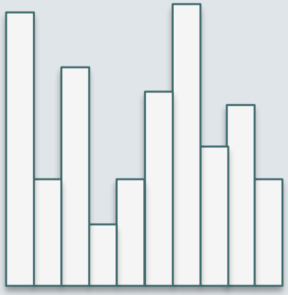
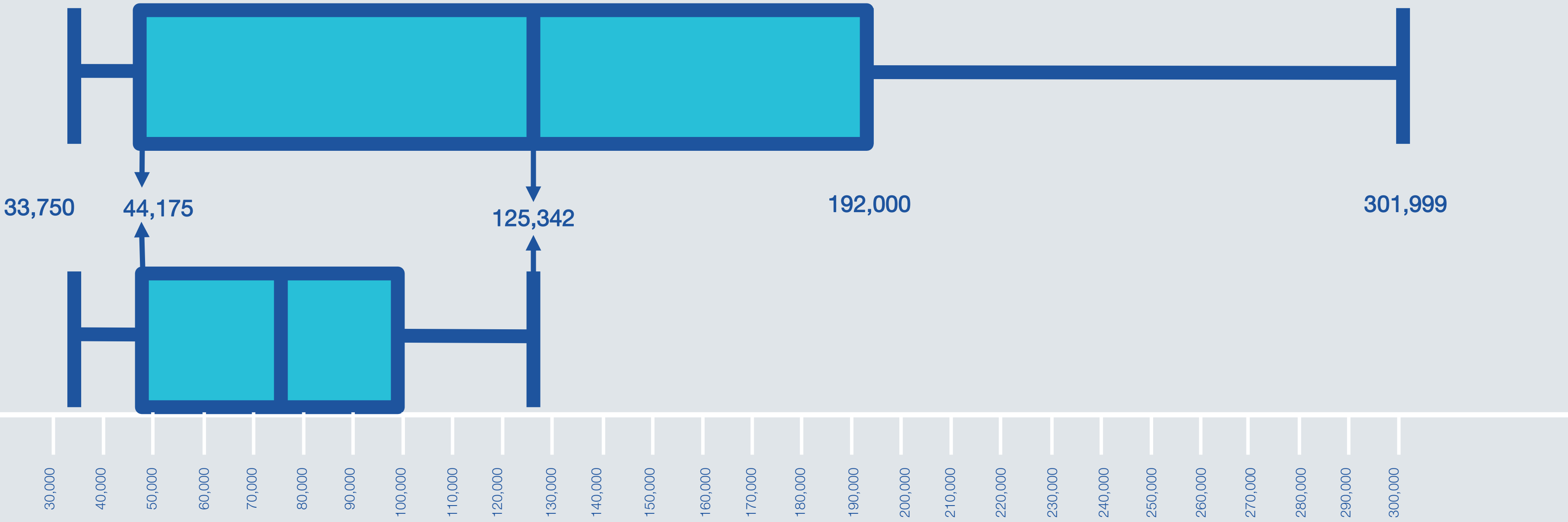


Multimodal distribution

(many modes)

Box plots are excellent for comparing datasets



Train strike: How much are rail workers paid?

7 December 2022

In a debate about the strikes in Parliament on 15 June, then Transport Secretary Grant Shapps said: "The median salary for a train driver is £59,000, compared with £31,000 for a nurse and £21,000 for a care worker."

The trouble with using the figure for train drivers in a debate about the RMT strikes is that the drivers are represented by their own union, Aslef, which is not taking part in the national strike.

https://www.bbc.co.uk/news/61840077

To work out the median salary, if you put all train drivers in a row in order of their pay, the person in the middle of the row would be earning the median.

Mr Shapps was mostly right on these figures, which came from the **Office for National Statistics** (ONS).

Mr Shapps went on to say: "The median salary for the rail sector is £44,000, which is significantly above the median salary in the country."

We asked the Department for Transport (DfT) how it got to this figure and it initially said it had taken the median figures from the ONS for four categories of rail workers, added them up and divided by four:

This wouldn't have produced a median figure and the DfT subsequently got in touch to say they had actually got the figure **from the ONS**, which had produced a genuine median figure (£43,747) across everybody in those four categories, along with a fifth - workers who build and repair engines and carriages (their median salary was £46,753).

https://www.bbc.co.uk/news/61840077

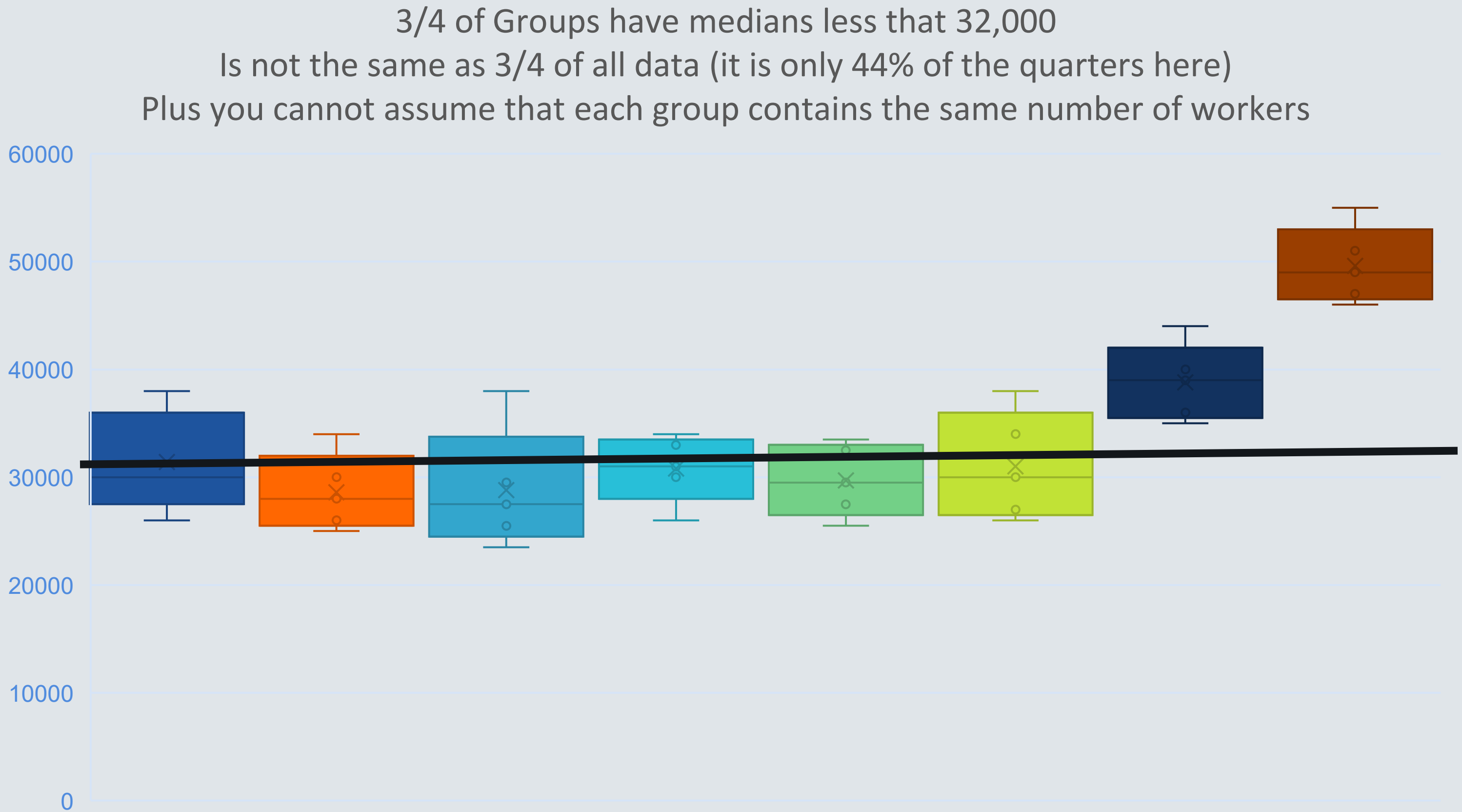
RMT general secretary Mick Lynch told BBC News on 6 December: "75% of the people in this dispute are paid less than £32,000 a year for working 24/7 shifts."

We asked the union how this had been worked out. It sent us a list dividing 33,000 members involved in the dispute into eight categories of workers, and giving the median salary for each of those categories.

The list shows that three quarters of the workers are in categories for which the median salary is below £32,000, but that means some of them could be earning more than £32,000.

https://www.bbc.co.uk/news/61840077

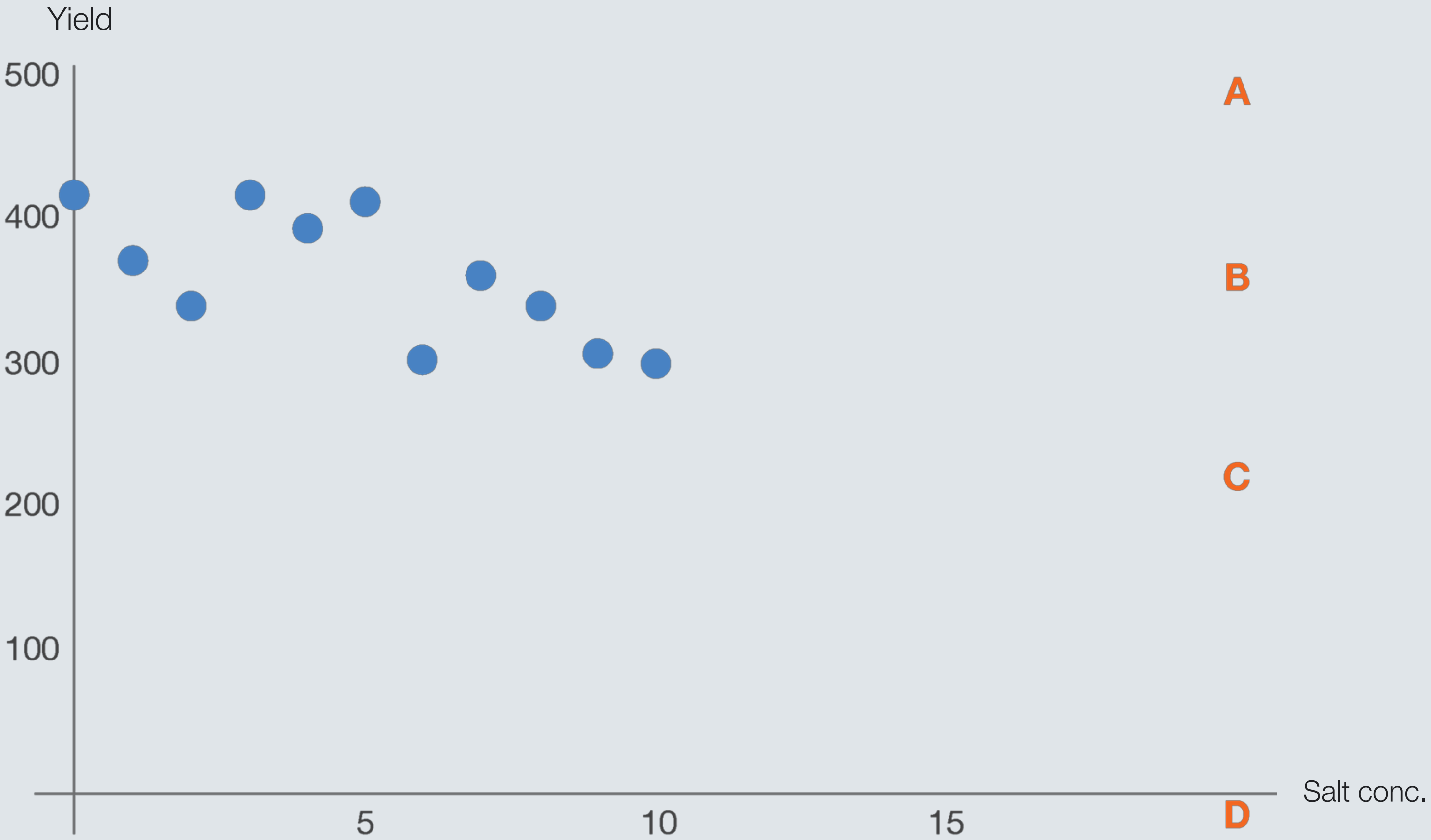
75% of workers are paid less than 32,000?



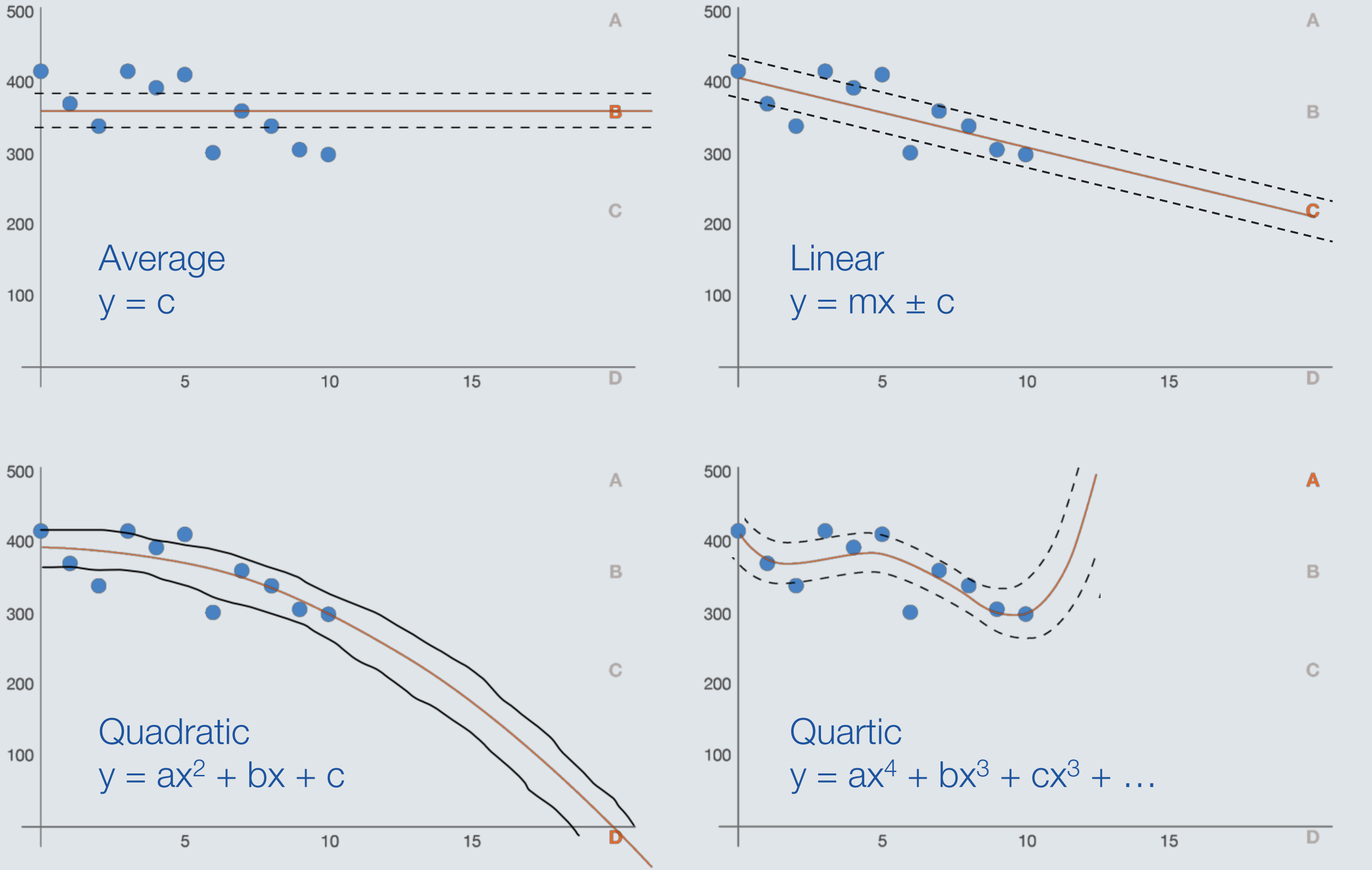
* Not real data!

Inferential statistics allows you to make predictions and classifications based on data.

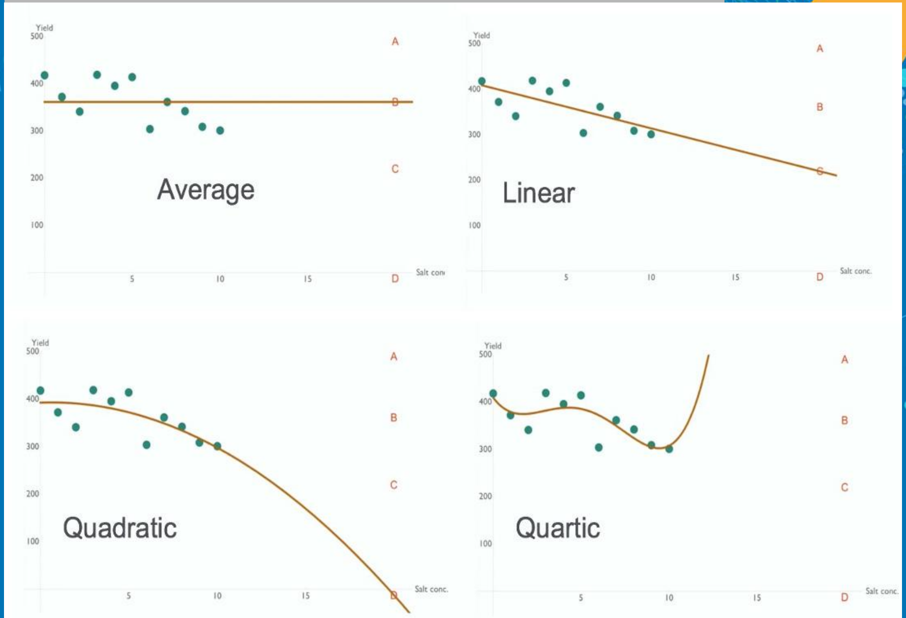
Trend Analysis



Overfitting



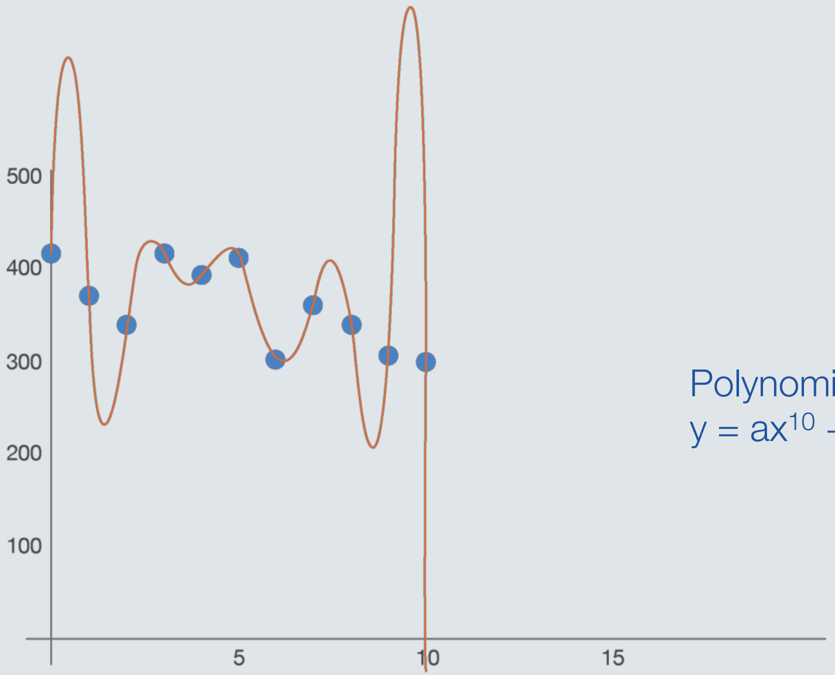
ML Algorithm #1



Regression

Used to establish the relationship between two independent variables by fitting a line of best fit.

Overfitting



Polynomial
$y = ax^{10} + bx^{9} + cx^{8} + \ldots$

A perfect model of the past, is a rubbish predictor of the future!

Descriptive statistics vs Inferential statistics

* not only will it be rubbish you will open yourselves up to accusations of discrimination where it goes wrong.

Overfitting

Knowing which trade off is acceptable is key

Objective

To develop an image recognition system that can recognise cattle

True positive: Algorithm says: "Is Cattle" is correct	**False positive:** Algorithm says: "Is Cattle" is wrong
True negative: Algorithm says: "Not Cattle" is correct	**False negative:** Algorithm says: "Not Cattle" is wrong

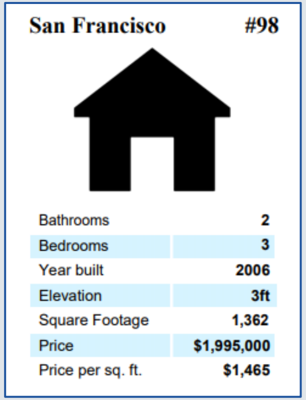
Knowing which trade off is acceptable is key

Objective

To develop an AI system that can diagnose cancer

True positive: Algorithm says: **"Has cancer"** **is correct** - patient has cancer	**False positive:** Algorithm says: **"Has cancer"** **is wrong** - patient does not has cancer
True negative: Algorithm says: **"No cancer"** **is correct** - patient has not got cancer	**False negative:** Algorithm says: **"No cancer"** **is wrong** - patient has got cancer

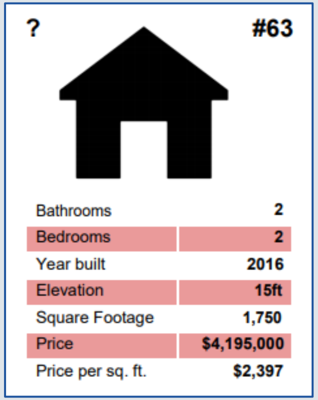
Activity 2



San Francisco	#98
Bathrooms	2
Bedrooms	3
Year built	2006
Elevation	3ft
Square Footage	1,362
Price	$1,995,000
Price per sq. ft.	$1,465

Each group has a set of “Top Trump” cards relating to properties in two cities.

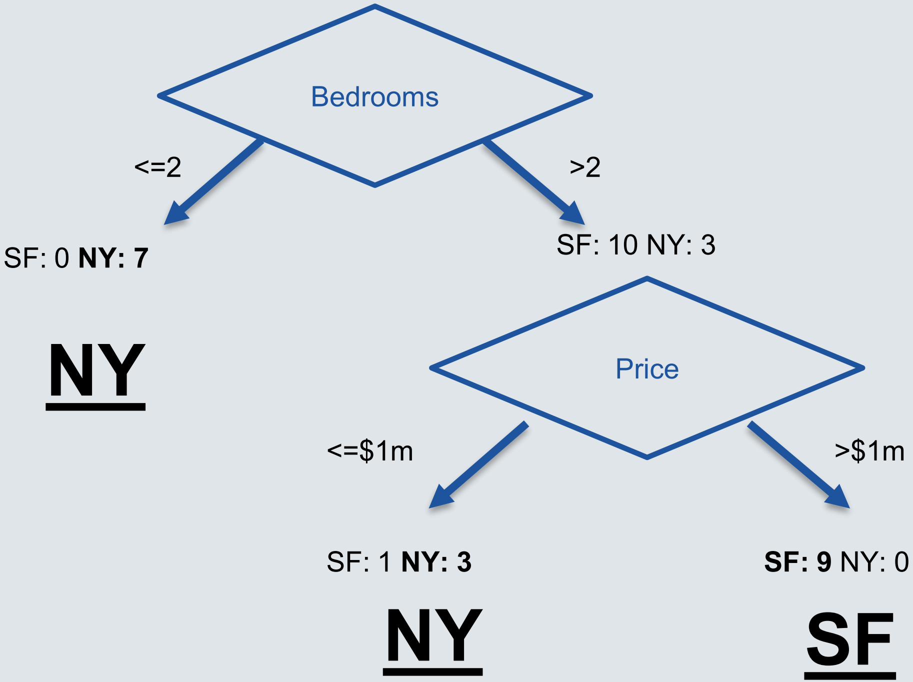
Your challenge is to build a decision tree that can predict the city based upon just the characteristics.



?	#63
Bathrooms	2
Bedrooms	2
Year built	2016
Elevation	15ft
Square Footage	1,750
Price	$4,195,000
Price per sq. ft.	$2,397

Example decision tree

95%
Confidence



Correct: 19
Wrong: 1

What outcome did you focus on?

1. I'm happy as I have separated all the New Yorks from all the San Francisco in the training data.

2. I was trying to achieve a complete separation of New York and San Francisco but didn't get there in the time

3. I have a model which doesn't completely separate New York and San Francisco in the training data and that is what I intended.

Approaches

Data first

Knowledge first

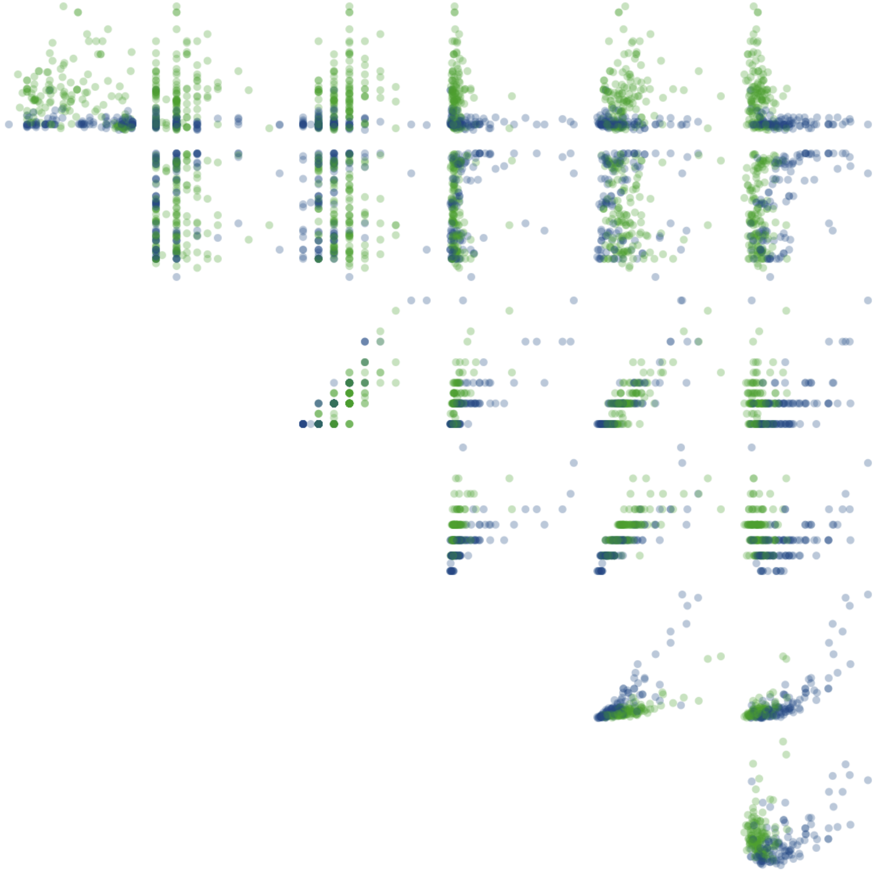
Many thanks to Tony and Stephanie for the inspiration and dataset behind the ML Card Game.

A visual introduction to machine learning

r2d3.us

Part 1: A Decision Tree

Part 2: Bias and Variance

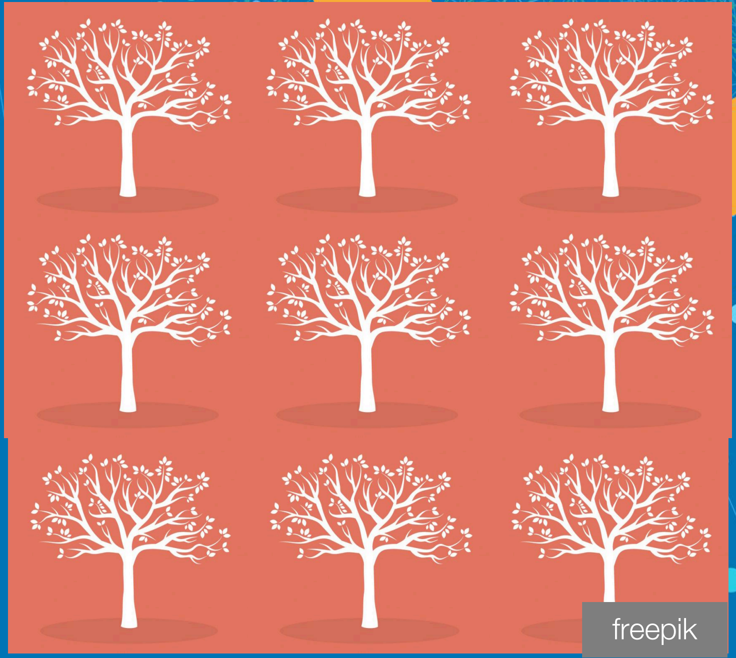


ML Algorithm #2

Decision trees

A decision tree is a uses a tree-like graph or model of decisions and their possible consequences, including chance event outcomes, resource costs, and utility. It is one way to display an algorithm that only contains conditional control statements.

ML Algorithm #3



Random Forests

To classify a new object based on attributes, each tree gives a classification and we say the tree “votes” for that class. The forest chooses the classification having the most votes (over all the trees in the forest) and in case of regression, it takes the average of outputs by different trees.

Card No.	Group #1	Group #2	Group #3
#23	SF	…	…
#59	NY		
#233	SF		

Algorithm summaries

TYPE	NAME	DESCRIPTION	ADVANTAGES	DISADVANTAGES
Linear	Linear regression	**The "best fit" line** through all data points. Predictions are numerical.	Easy to understand -- you clearly see what the biggest drivers of the model are.	X Sometimes too simple to capture complex relationships between variables. X Tendency for the model to "overfit".
	Logistic regression	The adaptation of **linear regression** to problems of classification (e.g., yes/no questions, groups, etc.)	Also easy to understand.	X Sometimes too simple to capture complex relationships between variables. X Tendency for the model to "overfit".

Algorithm summaries

Tree-based

Decision tree	A graph that uses a **branching method** to match all possible outcomes of a decision.	Easy to understand and implement.	X Not often used on its own for prediction because it's also often too simple and not powerful enough for complex data.
Random Forest	Takes the average of many decision trees, each of which is made with a sample of the data. Each tree is weaker than a full decision tree, but **by combining them we get better overall performance.**	A sort of "wisdom of the crowd". Tends to result in very high quality models. Fast to train.	X Can be slow to output predictions relative to other algorithms. X Not easy to understand predictions.
Gradient Boosting	Uses even weaker decision trees, that are increasingly **focused on "hard" examples.**	High-performing.	X A small change in the feature set or training set can create radical changes in the model. X Not easy to understand predictions.

Algorithm summaries

Neural networks

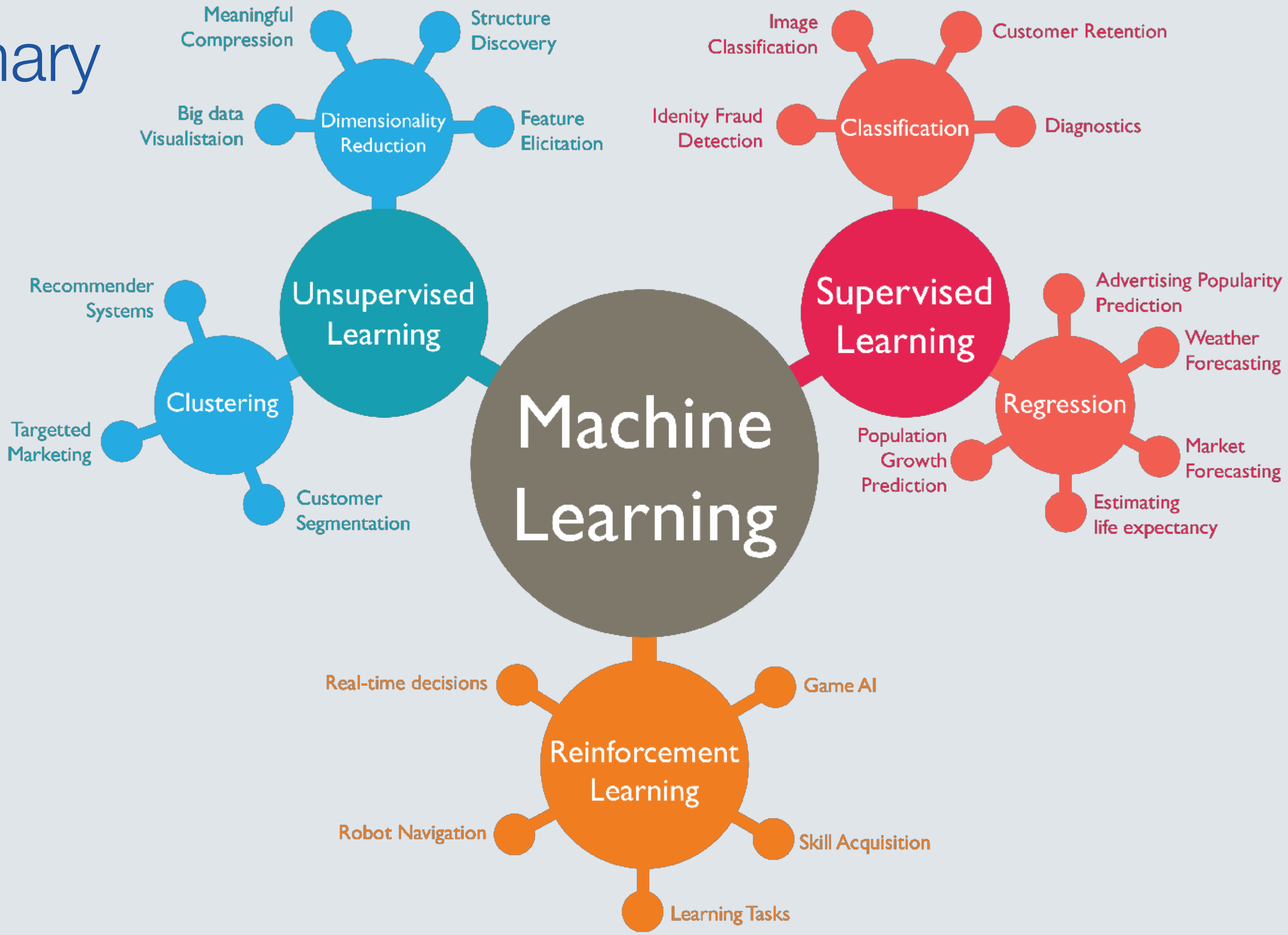
Neural networks

Mimics the behavior of the brain. Neural networks are interconnected neurons that pass messages to each other. Deep learning uses several **layers of neural networks put one after the other.**

Can handle extremely complex tasks - no other algorithm comes close in image recognition.

- X Very, very slow to train, because they have so many layers. Require a lot of power.
- X Almost impossible to understand predictions.

Summary


Machine Learning
Unsupervised Learning
Dimensionality Reduction
Meaningful Compression
Structure Discovery
Big data Visualistaion
Feature Elicitation
Clustering
Recommender Systems
Targetted Marketing
Customer Segmentation
Supervised Learning
Classification
Image Classification
Customer Retention
Idenity Fraud Detection
Diagnostics
Regression
Advertising Popularity Prediction
Weather Forecasting
Population Growth Prediction
Market Forecasting
Estimating life expectancy
Reinforcement Learning
Real-time decisions
Game AI
Robot Navigation
Skill Acquisition
Learning Tasks

Is it possible to be fair?

What does fairness mean? Fair to whom?

#1: You are looking to gain insights into the satisfaction of a beauty product to understand how it is viewed against competitors. Which of the following would you say is the best source for reliable data?

1. Collecting social media data from 10,000 people
2. A survey of 1000 people on the high street
3. A survey of 100 customers across the country
4. A survey of 50 beauty professionals
5. Other?

#2: You want to work out which routes people take to get between two stations on the London Underground where there is no direct route.

How would you do this?

You cannot use ticket validations or tap in/out as this only tells us a person travelled between two locations; it does not help us identify the route they used.

#3 You want to ensure good DEI representation on a TV Chat show.

How would you measure it?

#4 You want to implement predictive policing.

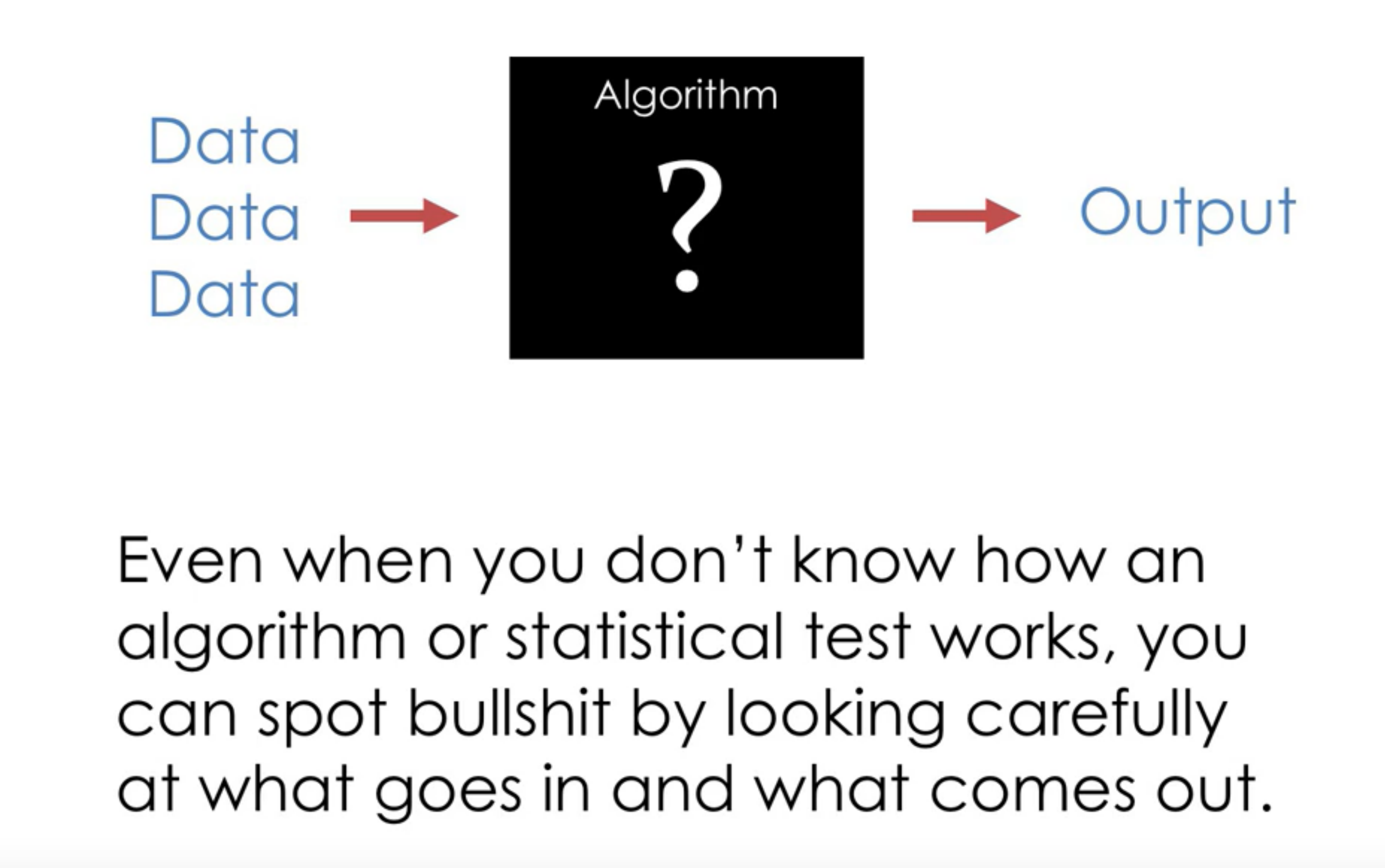
What data can help you do this and what limitations do you need to be aware of?

#5 You want to gather data about people so you can quote them for car insurance.

What data points do you need to work out risk?
What data points are only needed for the contract?

#6 You want to gather data points about a convicted individual to ascertain their reoffending risk.

What data points would you collect?
What would you not collect?



https://www.youtube.com/watch?v=_AXyeKbw3tU#t=9m23s

Calculating recidivism risk

Prisoners given questionnaire by trained prison staff.

Each question earns 1 point if negative.

Prisoners are determined to be high, medium or low risk for recidivism–this score is presented to a judge at sentencing.

Research found that only 11 of the questions had a statistical associate with recidivism.

Variable	% Agreement	Variable	% Agreement
I. Criminal History		**V. Accommodations**	
1. Any prior convictions?	96	27. Unsatisfactory	**64**
2. Two or more prior convictions?	93	28. Three or more address changes last year?	**82**
3. Three or more convictions?	93	29. High crime neighborhood?	**67**
4. Three or more present offenses?	**81**	**VI. Leisure/Recreation**	
5. Arrested under age 16?	**78**	30. Absence of recent participation in an org. activity?	**53**
6. Ever incarcerated upon conviction?	95	31. Could make better use of time	**53**
7. Escape history from a correctional facility?	**81**	**VII. Companions**	
8. Ever punished for institutional misconduct?	**87**	32. A social isolate?	**68**
9. Charge made for probation/parole suspended during prior community supervision?	91	33. Some criminal acquaintances?	**77**
10. Official record of assault/violence?	**86**	34. Some criminal friends?	**59**
II. Education/Employment		35. Few anti-criminal acquaintances?	**61**
11. Currently employed?	**86**	36. Few anti-criminal friends?	**59**
12. Frequently unemployed?	**72**	**VII. Alcohol/Drug Problem**	
13. Never employed for a full year?	**72**	37. Alcohol problem, ever?	**76**
14. Ever fired?	**78**	38. Drug problem, ever?	**88**
15. Less than regular grade 10?	**85**	39. Alcohol problem, currently?	**72**
16. Less than regular grade 12?	**88**	40. Drug problem, currently?	**68**
17. Suspended or expelled at least once?	**76**	41. Law violations?	**79**
18. Participation/performance	**78**	42. Marital/family?	**68**
19. Peer interactions	**76**	43. School/work?	**62**
20. Authority interactions	**75**	44. Medical?	**78**
III. Financial		45. Other indicators?	**63**
21. Problems?	**60**	**IX. Emotional/Personal**	
22. Reliance upon social assistance?	**69**	46. Moderate interference?	**84**
IV. Family/Marital		47. Severe interference, active psychosis?	**93**
23. Dissatisfaction with marital or equivalent situation?	**64**	48. Mental health treatment, past?	**87**
24. Non-rewarding, parental	**62**	49. Mental health treatment, present?	**89**
25 Non-rewarding, other relative	**65**	50. Psychological assessment indicated?	**66**
26. Criminal family/spouse?	**68**	**X. Attitudes/Orientation**	
		51. Supportive of crime	**68**
		52. Unfavorable toward convention	**78**
		53. Poor, toward sentence?	**72**
		54. Poor, toward supervision?	**62**

Calculating feature importance

So one way to give a statisatical measure to feature importance is to calculate the **Gini Impurity** of each factor.

We can first calculate the impurity before splitting (e.g. the purity of the whole dataset). So here impurity if you considered no factors. Given we have an even amount of data in each target class (NY and SF) then logically the impurity is 0.5:

$$G_{before} = 1 - (p_{SF}^{2} + p_{NY}^{2})$$

$$G_{before} = 1 - \left(\left(\frac{10}{20}\right)^2 + \left(\frac{10}{20}\right)^2\right) = 0.5$$

Next, lets split them by elevation of 12.

	NY	SF	Gini impurity per group
< 12	8	2	$1 - \left(\left(\frac{8}{2+8}\right)^2 + \left(\frac{2}{2+8}\right)^2\right) = 0.32$
>= 12	2	8	$1 - \left(\left(\frac{2}{2+8}\right)^2 + \left(\frac{8}{2+8}\right)^2\right) = 0.32$

Now we have the impurity for each leaf we will calculate the weighted average for elevation, which gives us an overall impurity for the factor. Note this takes into consideration if the dataset is not weighted equally between target labels.

For our example this will be: $\left(\frac{10}{20} * 0.32\right) + \left(\frac{10}{20} * 0.32\right) = 0.32$

Feature importance on our Houses

When dealing with **continuous features** like **elevation**, you need to determine an **optimal split point** to classify the data effectively. This is done by testing potential boundaries and selecting the one that leads to the greatest reduction in impurity.

The table here shows some example boundaries and their impurity.

Feature importance is calculated by subtracting the new impurity from the impurity before the split. So higher is better!

However, this is only a suggestion of which features are significant. It doesn't tell us which are most accurate or fair, which might require different boundaries.

Feature	Threshold	Gini Impurity	Feature importance (G_{before} - G_{after})
price_per_sqft	875.00	0.167	0.333
beds	1.50	0.313	0.187
elevation	12.00	0.32	0.18
bath	1.25	0.374	0.126
sqft	860.00	0.405	0.095
year_built	2007.50	0.444	0.056
price	2,085,000.00	0.469	0.031

Apple card case study

In this case study, we aim to replicate the work done in a white paper , co-authored by *Microsoft* and *EY*, on mitigating gender-related performance disparities in financial lending decisions. In their analysis, Microsoft and EY demonstrated how Fairlearn

could be used to measure and mitigate unfairness in the loan adjudication process.

Because the dataset used in the white paper is not publicly available, we will be making use of a publicly available dataset and following the methodology outlined in this Fairlearn notebook.

Apple Card is accused of gender bias. Here's how that can happen

By Evelina Nedlund, CNN Business

5 minute read · Updated 2:04 PM EST, Tue November 12, 2019

Steve Wozniak
@stevewoz

I'm a current Apple employee and founder of the company and the same thing happened to us (10x) despite not having any separate assets or accounts. Some say the blame is on Goldman Sachs but the way Apple is attached, they should share responsibility.

7:06 am · 10 Nov 2019

Where did the bias originate?

Training data:

40% male
60% female

22% defaulted on their loans

43% male
57% female

The dataset was rebalanced so the target class (defaulted) is not overpowered by features from the majority class.

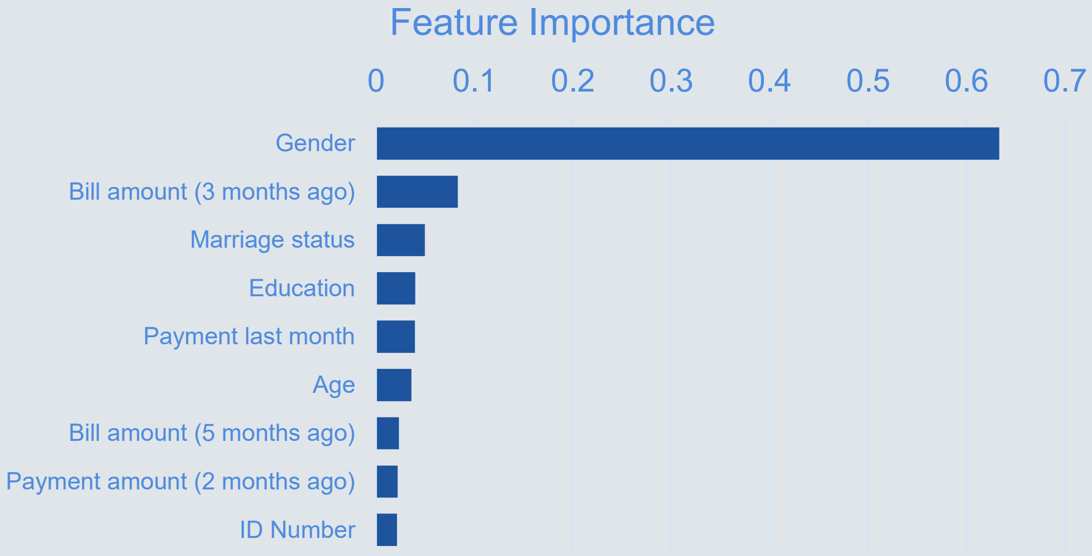
50% defaulted
50% didn’t default

Looking at the MS/EY case study

The use case from Microsoft and EY shows the following graphic (we have removed a derived feature and re-balanced) to illustrate the characteristics their model used to build the predictions.

We can see gender is the main influencer for how the model will predict if a person will default on their payment, and by a significant margin. No other feature comes anywhere near as close to predicting if a person will default, so you can see why a credit card company might want to include gender in their algorithm.

Interestingly, the amount paid last month is the 5th best indicator, with the second best apparently being the bill amount from 3 months ago; do you think this is highlighting a meaningful underlying pattern in the data, or is it correlation without causation?



Calculating balanced accuracy

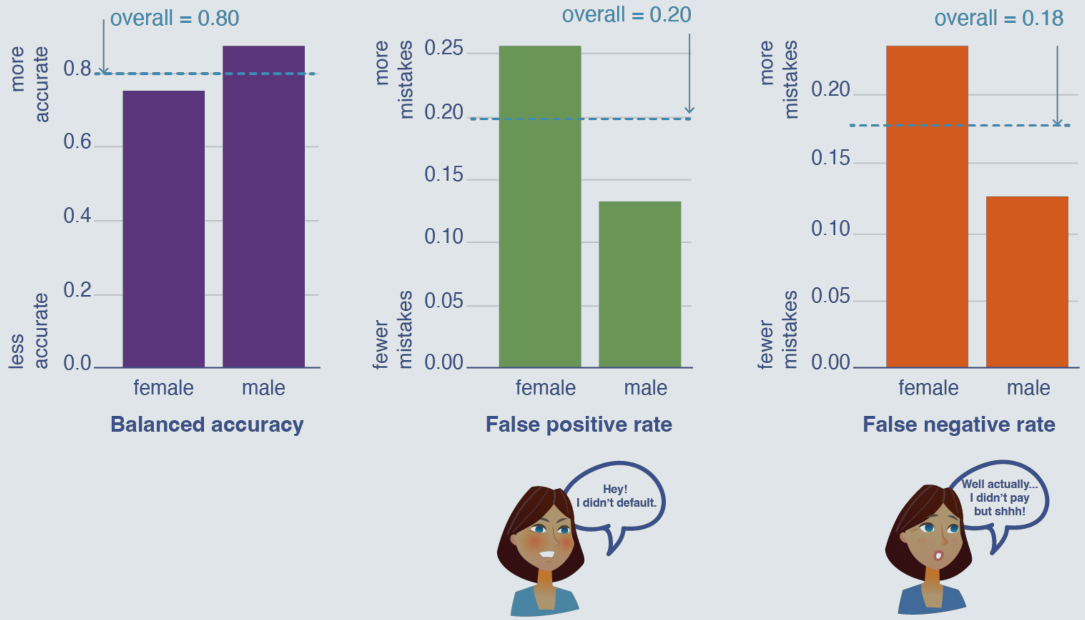
$$Accuracy = \frac{1}{2}\left(\frac{no.\,of\ people\ who\ actually\ defaulted}{total\ predicted\ to\ default} + \frac{no.\,of\ people\ who\ actually\ didn't\ default}{total\ predicted\ to\ not\ default}\right)$$

A balanced accuracy looks at the model's ability to correctly classify people into both categories –so, as either going to default or not. It is worked out using an equation that makes use of the True Positive, True Negative, False Positive and False Negative rates

Balanced accuracy is a calculation of the probability of correct classifications for both cases, and it then takes the average to give the overall score of the model's ability to do the job.

$$Accuracy = \frac{1}{2}\left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP}\right)$$

Original model



But this doesn’t mean it is fair, as it will achieve the objective by discriminate?

If the company offers loans to people who are not able to repay them, the business will be negatively impacted.

To minimise this impact, which metric should be focussed on?

True positive: Algorithm says: “Will default” is correct	**False positive:** Algorithm says: “Will default” is wrong – customer paid
True negative: Algorithm says: “Won’t default” is correct – paid	**False negative:** Algorithm says: “Won’t default” is wrong – defaulted

But this doesn’t mean it is fair, as it will achieve the objective by discriminate?

If a company wants to minimise the harm caused from predicting someone cannot pay their loan, when in fact they can, which metric should be focussed on?

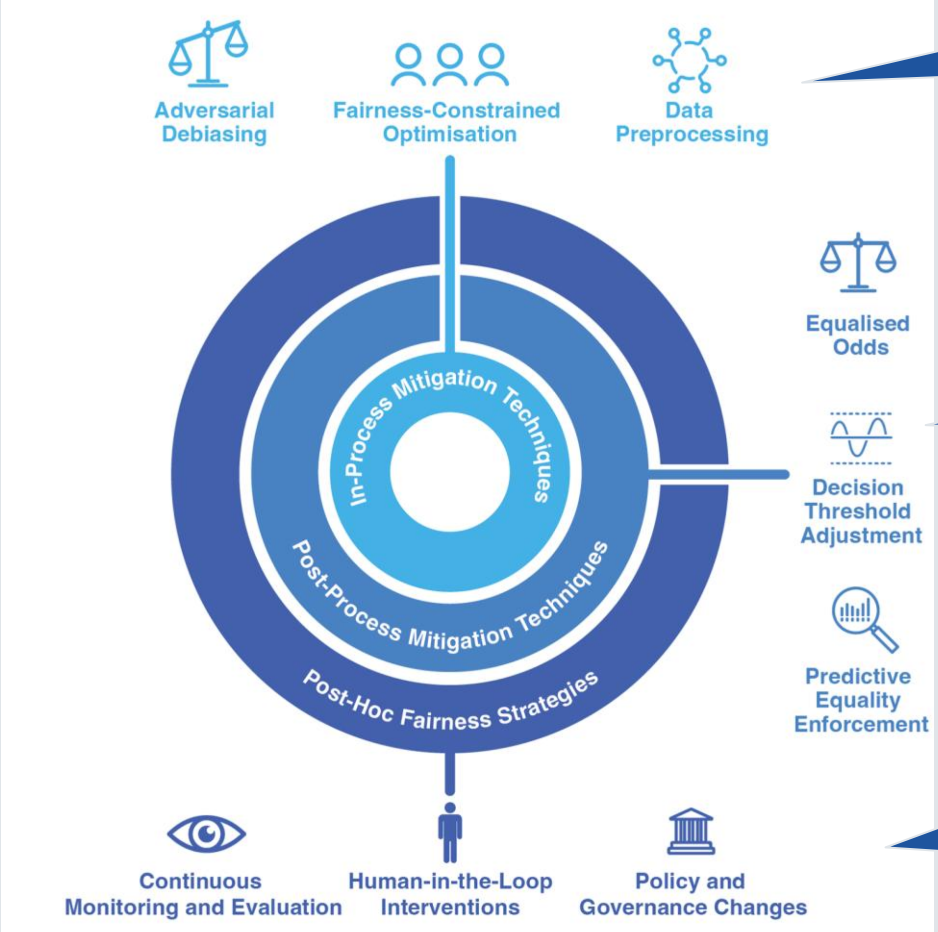
True positive: Algorithm says: “Will default” is correct	**False positive:** Algorithm says: “Will default” **is wrong –** customer paid
True negative: Algorithm says: “Won’t default” **is correct –** paid	**False negative:** Algorithm says: “Won’t default” **is wrong –** defaulted

Is it possible to be fair?

What does fairness mean? Fair to whom?

Took a long time to get here I know! Sorry

When can bias be addressed?



In-process mitigation techniques
applied during training phase

Post-process mitigation techniques
adjustments to the outputs

Post-hoc fairness strategies
do not need to be technical; don't change model outputs, but may change how they are applied

Fairness constrained optimisation

There are many different approaches to creating a fairness-aware algorithm.

One effective method is to adjust the **loss function** by applying weights to different types of errors. The idea here is to penalize the algorithm more for making incorrect predictions for one group—in this case, females—compared to another group, such as males.

Equal loss (original model)

Females wrong	Males wrong	Female loss weight	Male loss weight	Total loss
3	2	1	1	5

Weighted loss (original model)

Females wrong	Males wrong	Female loss weight	Male loss weight	Total loss
3	2	1.5	1	6.5

Weighted loss (new model)

Females wrong	Males wrong	Female loss weight	Male loss weight	Total loss
1	4	1.5	1	5.5

Equalised Odds

Equalised odds is applied after the model has been trained.

Typically, a model doesn't output a hard "yes" or "no" but rather a continuous score or probability that an instance belongs to the positive class.

For a decision tree, when a leaf is formed, it contains several samples with various true class labels. Instead of simply assigning the majority class as the prediction, you can calculate the probability for each class by looking at the fraction of samples in that leaf that belong to that class. For instance, if a leaf node has 100 samples, and 80 of them belong to the positive class while 20 belong to the negative class, then the probability of a positive prediction would be 0.8 (or 80%). So we can get a prediction and confidence from a decision tree.

Threshold 0.5

Gender	Threshold	False positive rate
Male	0.5	0.13
Female	0.5	0.26

Equalised odds

Gender	Threshold	False positive rate
Male	0.55	0.27
Female	0.45	0.27

It should be noted this example is illustrative. We have not used the adjusted thresholds from the study.

Adjusting for fairness

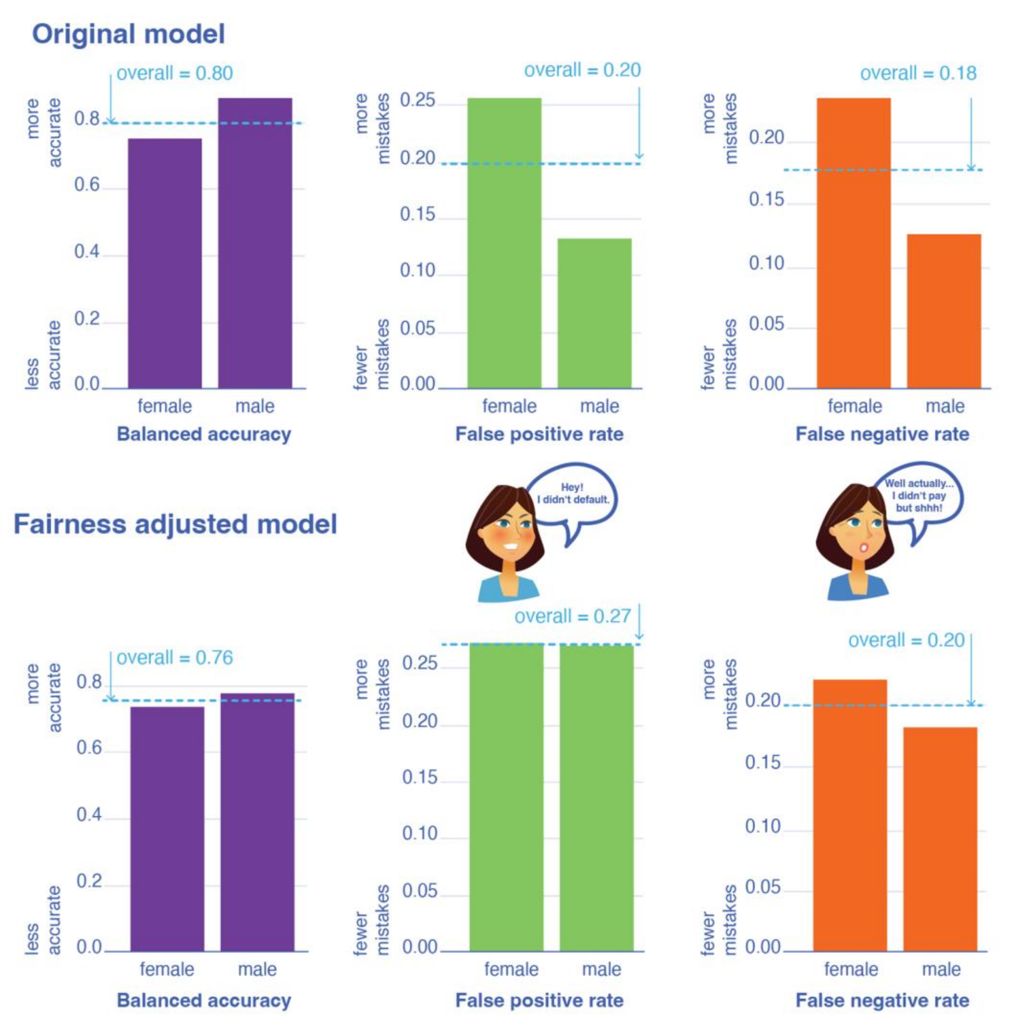
This approach adjusts the disparity found in the sensitive feature (gender) by adjusting the threshold of the classification criteria until the predictions meet a specific objective.

In this case, the objective was to reduce the bias in the balanced accuracy, so that the model is not making more mistakes with women than with men.

The fairness adjusted model resulted in an increase in the false positive and false negative rates for men.

Is that fair?

Who says?



Exploring trade-offs

Ensuring the algorithm is as fair as possible

Ensuring the algorithm is as fair as possible to demographic groups

Minimising processing of personal data

Ensuring the algorithm is as fair as possible to individuals

Achieving the most accurate possible prediction?

Having an algorithm that is easily understood and explained

Minimising computational impact (financial & environmental)

Is AI explainable?

That’s all the technical stuff, can you explain it to someone else?

If you can’t should we make it that complicated?

Terminology

XAI:
Explainable AI - enabling people to understand (and therefore trust) results & outputs of AI systems

Interpretability: documentation or tools to help stakeholders interpret results & understand personal impact

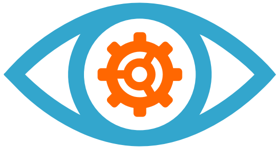
Accuracy: checkpoints for monitoring its ability to predict

Accountability: defining roles & responsibilities - what happens if discriminatory patterns emerge?

Fairness: checkpoints for monitoring discriminatory patterns - ensuring safeguards against bias

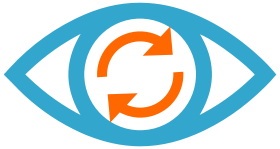
Transparency approaches:

which are most important who various stakeholders? In which sectors?



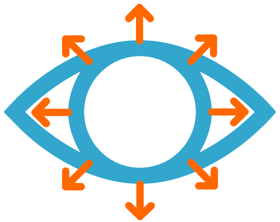
Technical transparency

Training data
Model architecture
Algorithms



Process transparency

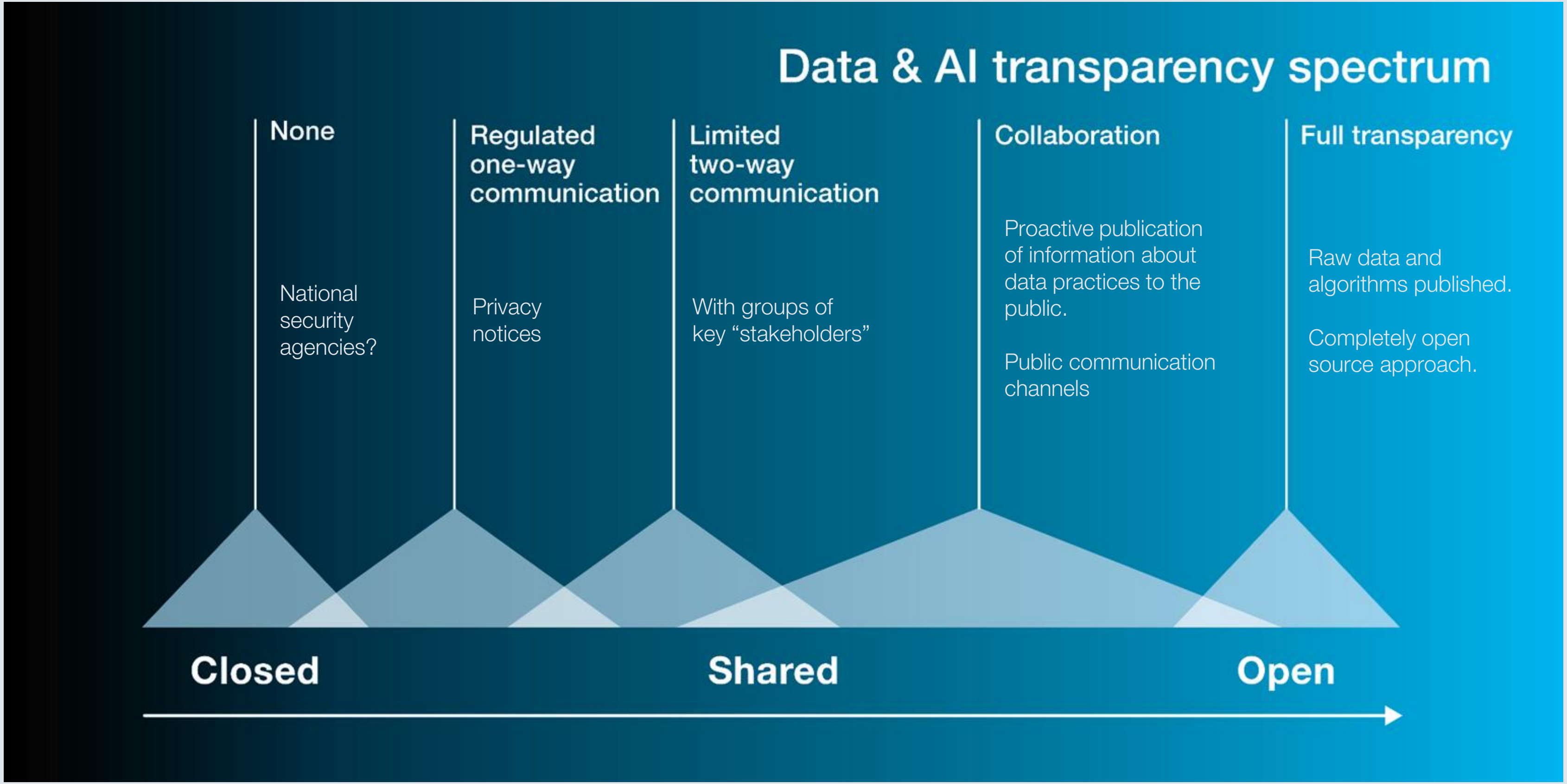
Documentation of development
Information about decision-making
How will system be used?
Audits



Outcome transparency

Information about performance for different demographic groups
Explaining specific decisions
Allowing challenges/disputes

Data and AI Transparency Spectrum



IEEE: P7001 Standard for Transparency of Autonomous Systems

The IEEE P7001 Standard for Transparency of Autonomous Systems was developed by a diverse working group comprising hundreds of participants from academia, industry, civil society, policy, and government across six continents.

The working group established five principles for Ethically Aligned Design, one of which is Transparency.

The IEEE P7001 Standard for Transparency of Autonomous Systems outlines five distinct types of stakeholders and assigns six levels of transparency for each, ranging from Level 0 to Level 5.

P7001 Stakeholder group	Bare minimum (level 1)	Highest access (level 5)
Users	written/visual/audio explanation of how the system behaves	Access to log files and training data
General public	Visual indicator that the service employs autonomous systems (basic awareness)	Stakeholders can make data governance requests to companies
Certification agency	Documentation explaining how the system works & what decisions it makes	Access to source code, statistical models, training data, info on data collection methods
Incident investigator	Access to video/audio records for basic analysis	Access to comprehensive data and documentation for thorough audits
Expert advisor (legal / admin)	Certifications that system has passed audits	Detailed records of certifications, audits, incident reports, documentation of actions taken

AI4People's Ethical Guidelines

The AI Ethical Guidelines aim to provide guidance on ethical considerations in artificial intelligence (AI).

These guidelines were developed by researching principles from various existing ethics frameworks, identifying commonalities among them, and grouping the 47 found principles into four categories consistent with previous ethics work.

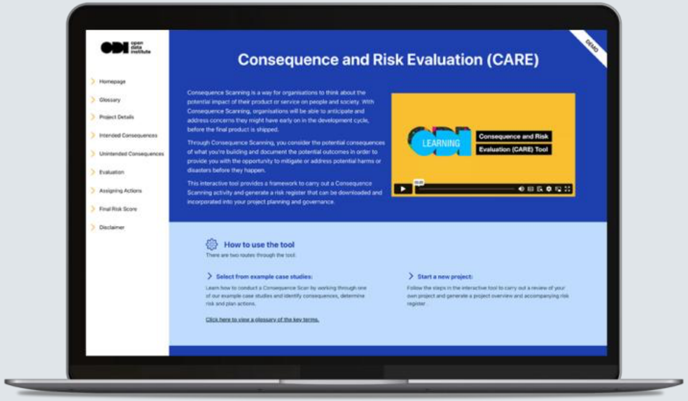
The novel addition is the "Explicability" principle, emphasising the need for transparency in AI systems.

Specific to Explicability, the guidelines suggest developing industry-specific frameworks to enhance AI system transparency and provide clear explanations of decision-making processes, especially in cases of unintended consequences.

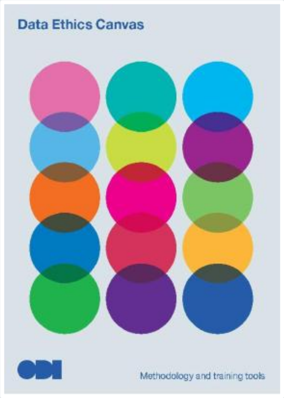
Data Ethics Tools

Project Level

Consequence scanning
Quick assessment

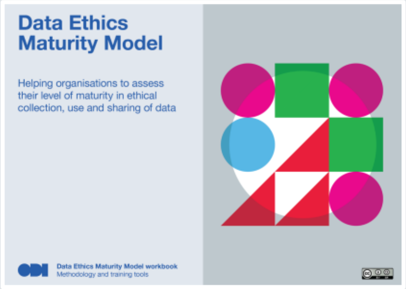


Consequence and Risk Evaluation (CARE)
Interactive framework and risk register



Data Ethics Canvas
Deep consideration

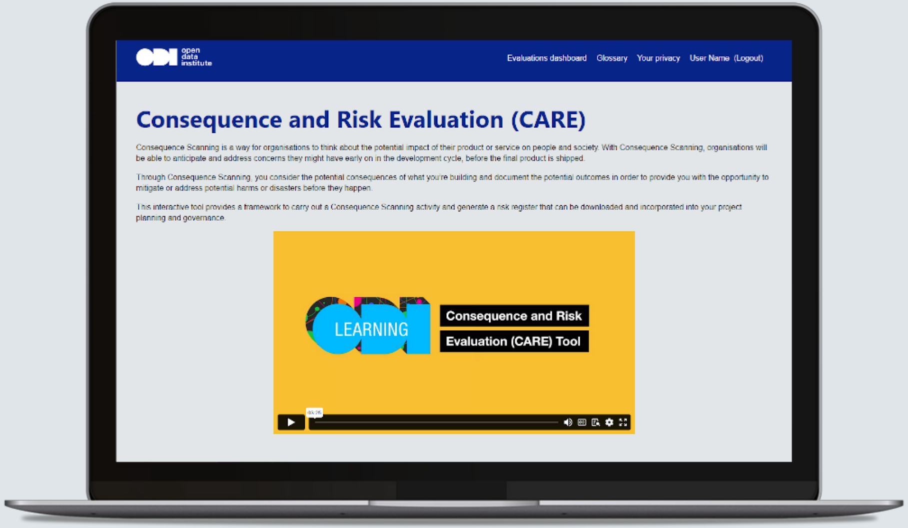
Organisational Level



Data Ethics Maturity Model
Benchmarking assessment

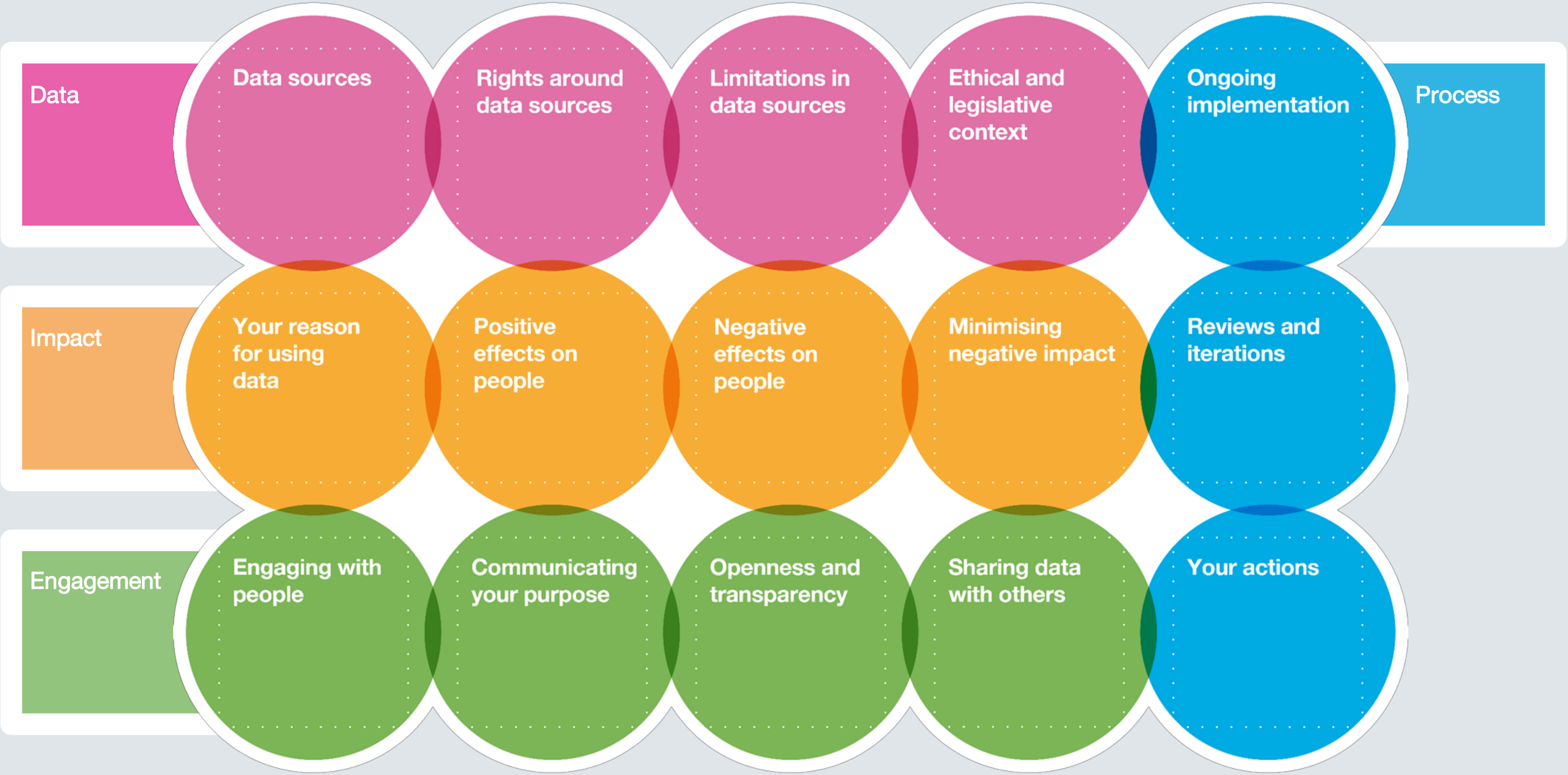
Consequence and Risk Evaluation Tool (CARE)

The **Consequence and Risk Evaluation** (CARE) Tool helps organisations assess and mitigate the societal impact of their products and services. It facilitates structured consequence scanning, focusing on both positive outcomes and potential risks to guide responsible innovation.



- ✓ Proactive risk identification and mitigation
- ✓ AI-driven insights to enhance assessments
- ✓ Downloadable risk reports and data exports

Data Ethics Canvas



The Data Ethics Canvas intends to help you:

Assess the data

By asking you to assess the data sources, their limitations, the rights and legislative context of data collection, use, and share that apply to your sector.

Identify the impact

By asking you to identify your reasons for using data to develop this service/product, the potential positive and negative impact on people and society, and the potential approaches to minimise negative harm.

Evaluate the engagement

Challenges you to think about the people involved, your approaches to communication and transparency

Decide the process

Helps you decide on the next steps: what should you prioritise, when will you iterate, do you need training, etc

Strategy and Ethics

Data Ethics Essentials

Develop an understanding of data ethics as a domain.

Explore the importance of data ethics practices through real-world examples.

Reflect on your own professional practice in relation to data and data ethics.

Apply tools and frameworks to support ethical thinking around the collection, sharing and use of data.

https://learning.theodi.org/courses/data-ethics-essentials

Cost: £FREE

Strategic Data Skills

This course will equip you with the theoretical knowledge, practical skills and technical know-how to make better decisions using data. It's a comprehensive course for people who want to work strategically with data, but don't need or want to learn complex programming.

Tutor Led

Self-Paced

Certified Data Ethics Professionals

Become a certified data ethics professional.

In this 60 hour course you will learn the knowledge, skills, and ability to apply tools and educate others on how to use, collect and share data responsibly, minimising harmful impacts. Modules include what is data ethics; ethics and personal data; working with data from 3rd party sources; data analysis and ethics in practice and operationalising data ethics.

Understand data ethics and AI (self-paced)

The Foundations of Ethical Decision Making

Explore different ethical frameworks, their strengths, their weaknesses and application to AI Learn how to navigate ethical challenges through a principled step by step SAD-AM model. Explore the inescapable presence of bias and break the boundary between ethics and law. Ensure inclusivity, ethical growth and organisational impact.

Legal and Regulatory Frameworks

Learn about the pivotal regulations that exert influence across all phases of the data lifecycle, with a particular emphasis on emerging AI regulations. Understanding the legal and regulatory frameworks that govern data practices to ensure compliance. Understand the purpose of data licences, interoperability and the importance of transparency to maintain trust.

Ethics and Personal Data

Develop the skills to handle personal data responsibly. Investigate data anonymisation techniques: how to mitigate the risks whilst maintaining the utility of data. Gain hands-on experience in handling sensitive information and engage with our AI assistant tutor for instant feedback on your work.

Ethics in data collection

Explore ethical practices in the collection of data. Equip yourself with the knowledge and tools necessary to navigate the intricate terrain of data collection with ethics and integrity. Evaluate the sources of bias during data collection and mitigation strategies.

Bias and Fairness in Data and AI

Develop skill to ensure that algorithms that process data are "fair". Develop your understanding of bias in data and algorithms and actively identify instances of bias in real-world datasets. Uncover the impact and consequences of bias in data and artificial intelligence

Transparency and Explainability in Data

Explore key aspects of building trust in data practices and AI models. Implement techniques for developing AI models that are explainable and interpretable. Examine existing ethical guidelines and frameworks that promote transparency and explainability and how this can be embedded in your organisational context.

Tools and Frameworks in Evaluating Ethical Data Practices

Explore practical tools and frameworks to assess and enhance ethical data practices. Get hands-on experience with our Consequence and Risk Evaluation (CARE) tool and implement ethical data evaluation seamlessly into your workflows. Address the possibility of unintended consequences arising from AI deployment.

Thank you

Course materials:

https://training.theodi.org/ML101

For more information about how the ODI can support your data need:

- Consulting services
- Research and policy
- Membership
- Training

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